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libname datasets '/home/u64375008/';

PROC IMPORT DATAFILE= '/home/u64375008/G12.outcome.data.csv'
            OUT= datasets.hierarchy
            DBMS=CSV REPLACE;
            GETNAMES=YES;
            GUESSINGROWS=MAX;

RUN;

/*LMM*/
proc mixed data=datasets.hierarchy;
  class climate(ref = "Scenario 1") species(ref = "Conference") location(ref = "BE") ecotr_id tree_id;
  model quality_idx = species climate location / solution cl
    outpm=predmean outp=pred;
  random intercept / subject=tree_id(ecotr_id) type=un;
  random intercept / subject=ecotr_id type=un;
  estimate 'Scenario 2 vs Scenario 1' climate 1 0 0 -1 / cl;
  estimate 'Scenario 3 vs Scenario 1' climate 0 1 0 -1 / cl;
  estimate 'Scenario 4 vs Scenario 1' climate 0 0 1 -1 / cl;
run;

proc print data=predmean;
proc print data=pred;
run;

proc sgplot data=pred;
  vbox pred / category=climate
    fillattrs=(color=ligr)
    lineattrs=(thickness=2);
  yaxis label="Predicted Quality Index";
  xaxis label="Climate Scenario";
run;

PROC IMPORT DATAFILE= '/home/u64375008/quality_binary.csv'
            OUT= datasets.binary
            DBMS=CSV REPLACE;
            GETNAMES=YES;
            GUESSINGROWS=MAX;

RUN;

/*GEE*/
proc genmod data=datasets.binary;
  class climate(ref = "Scenario 1") species(ref = "Conference") location(ref = "BE") ecotr_id tree_id;
  model quality_b = species climate location / dist= binomial type3 link=logit;
  repeated subject = tree_id(ecotr_id) / type=exch covb corrw modelse;
  estimate 'Scenario 2 vs Scenario 1' climate 1 0 0 -1;
  estimate 'Scenario 3 vs Scenario 1' climate 0 1 0 -1;
  estimate 'Scenario 4 vs Scenario 1' climate 0 0 1 -1;
run;
/*IND give worse SE and wider CI*/
proc genmod data=datasets.binary;
  class climate(ref = "Scenario 1") species(ref = "Conference") location(ref = "BE") ecotr_id tree_id;
  model quality_b = species climate location / dist= binomial type3 link=logit;
  repeated subject = tree_id(ecotr_id) / type=ind covb corrw modelse;
  estimate 'Scenario 2 vs Scenario 1' climate 1 0 0 -1;
  estimate 'Scenario 3 vs Scenario 1' climate 0 1 0 -1;
  estimate 'Scenario 4 vs Scenario 1' climate 0 0 1 -1;
run;

/*GLM*/
proc glimmix data=datasets.binary;
  class climate(ref = "Scenario 1") species(ref = "Conference") location(ref = "BE") ecotr_id tree_id;
  model quality_b = species climate location / dist= binomial link=logit solution;
  random intercept / subject=tree_id(ecotr_id) type=un;
  random intercept / subject=ecotr_id type=un;
  estimate 'Scenario 2 vs Scenario 1' climate 1 0 0 -1 / cl;
  estimate 'Scenario 3 vs Scenario 1' climate 0 1 0 -1 / cl;

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estimate 'Scenario 4 vs Scenario 1' climate 0 0 1 -1 / cl;
run;

/*LMM Longitudinal*/
PROC IMPORT DATAFILE= '/home/u64375008/longitudinal_data.csv'
OUT= datasets.longdata
DBMS=CSV REPLACE;
GETNAMES=YES;
GUESSINGROWS=MAX;
RUN;

/* with dropout */
data datasets.longdata_clean;
set datasets.longdata;
quality_numeric = input(quality, ?? best32.);
time_numeric = input(time, ?? best32.);
drop quality time;
rename quality_numeric = quality time_numeric = time;
run;

/* without dropout */
PROC IMPORT DATAFILE= '/home/u64375008/time_data_clean.csv'
OUT= datasets.longitudinal
DBMS=CSV REPLACE;
GETNAMES=YES;
RUN;

proc mixed data=datasets.longdata_clean;
title 'No Removal';
class climate(ref = "Scenario 1") species(ref = "Conference") tree_id pear_id;
model quality = species climate time climate*time / solution;
random intercept / subject=tree_id type=un;
random intercept time / subject=pear_id(tree_id) type=un;
contrast 'Any climate scenario vs climate scenario 1'
climate*time 1 0 0 -1,
climate*time 0 1 0 -1,
climate*time 0 0 1 -1;
estimate 'climate scenario 2 - climate scenario 1' climate*time
1 0 0 -1;
estimate 'climate scenario 3 - climate scenario 1' climate*time
0 1 0 -1;
estimate 'climate scenario 4 - climate scenario 1' climate*time
0 0 1 -1;
run;

/* remove dropout */
proc mixed data=datasets.longitudinal;
title 'Remove Dropout';
class climate(ref = "Scenario 1") species(ref = "Conference") tree_id pear_id;
model quality = species climate time climate*time / solution;
random intercept / subject=tree_id type=un;
random intercept time / subject=pear_id(tree_id) type=un;
contrast 'Any climate scenario vs climate scenario 1'
climate*time 1 0 0 -1,
climate*time 0 1 0 -1,
climate*time 0 0 1 -1;
estimate 'climate scenario 2 - climate scenario 1' climate*time
1 0 0 -1;
estimate 'climate scenario 3 - climate scenario 1' climate*time

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